AOUANOUK Slimane

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Available worldwide for a 6-month internship starting February 2027.

Graduate Engineering student at Arts et Métiers and **Research Master's student in Robotics at Sorbonne Université**, focused on **motion planning and optimal control foundations**. Research experience in **Python/SciPy trajectory optimization**, real-time control, automated testing, and physical robot validation; applied **MATLAB/Simulink** to system identification and controller design.

EDUCATION — Combined GPA (CPGE + ENSAM): 3.92 / 4.00

Sorbonne Université — Paris, France **Sept 2026 – Aug 2027**
Research Master's (M2) in Robotics (ROB)
Selected current coursework: Nonlinear Control; State Estimation & System Identification (EKF, UKF); Robot Manipulators; Mobile Robotics (ROS 2, SLAM); Machine Learning (reinforcement learning, PyTorch).

Arts et Métiers Institute of Technology (ENSAM) — Paris, France **Sept 2024 – Jul 2027**
Combined BS × MS Engineering Degree — *Member of* ***ParisTech*** — **Ranked top 12%** (142 / 1223)
Relevant Coursework: Numerical Optimization, Robotics and Mechatronics, Experimental Mechanics and Validation Testing

WORK EXPERIENCE

Trajectory Optimization and Collision Avoidance for Autonomous Systems **Jan–Aug 2026**
Robotics Research Project & Internship — PIMM Lab / CNRS@CREATE; Paris (Jan–May) → Singapore (Jun–Aug) — Supervisor: Prof. Francisco Chinesta

- Co-developed **two-level navigation**: offline finite-horizon Euler–Lagrange-based **trajectory optimization in Python/SciPy (L-BFGS-B)** and online causal **Newton-based feedback control (Euler/Verlet)**
- Used **analytical gradients**, fixed endpoint constraints, and physical/safety penalties; benchmarked an **89.7% average nominal-planner runtime reduction** across three scenarios, with convergence tradeoffs
- Integrated **OpenCV/ArUco localization** and serial rover control, sustaining **19.84–19.86 Hz** median operation; physically validated **40/40 goal arrivals with zero collisions** across two frozen 20-run campaigns
- Co-developed **pytest unit and regression tests across 64 test files**, checking trajectory boundary conditions, obstacle avoidance, controller causality, and analytical gradients against finite differences
- **Explored Model Predictive Control (MPC) integration** for constrained receding-horizon control; the extension remained incomplete and was not validated on hardware

SELECTED PROJECTS

Experimental System Identification & Control of a Bicopter – CNAM **2026**
- Co-identified a second-order model from physical bicopter response data; designed and simulated a discrete **state-space controller with integral action and pole placement** in **MATLAB/Simulink**, then implemented it on **dSPACE hardware at 20 Hz**.

Vision-Language-Action Manipulation on a Franka Robot – Sorbonne Université **2026–27**
- Contributing across the full lifecycle of an ongoing **end-to-end VLA manipulation project** on a **Franka robotic arm**, covering **VR-teleoperated demonstration collection**, dataset preparation, **policy adaptation and post-training**, **ROS 2 integration on the robot**, real-robot deployment, and evaluation
- Developing a **multimodal VLA policy** that maps **language instructions and RGB/RGB-D observations** to continuous robot actions, with **openpi/π0** considered as a candidate foundation and evaluation focused on **task success, robustness, action latency, and human intervention**

SKILLS AND INTERESTS

Programming & Tools: Python (SciPy, L-BFGS-B), pytest, MATLAB/Simulink, Git, OpenCV; Linux (current coursework/tooling)
Numerical Optimization: Constrained optimization (fixed boundary conditions, physical-limit penalties), analytical gradients
Planning & Control: Trajectory optimization, optimal control foundations, dynamic modeling, real-time feedback control; MPC (integration explored)
Testing & Validation: Unit testing, regression testing, numerical validation, automated simulation sweeps, physical robot testing, experiment logging, technical documentation
AI-Assisted Development: Claude and Codex for code generation/review, debugging, documentation, and validation
Languages: French (native), English (C1; TOEFL iBT MyBest 5/6)
Interests: Autonomous navigation, robot learning, basketball, strength training